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HCF and LCM

HCF/ GCD
For a set of numbers, the HCF is the greatest # that divides all of them.

HCF $(1, z) = 1$

HCF $(x, y) = x$ if y is a multiple of x.

If HCF$(x, y) = 1$, they are coprime.

Division method:

56, 12

$$12 \overline{)56}\,(4 \quad -48 \quad 8 \overline{)12}\,(1 \quad -8 \quad 4 \overline{)8}\,(2 \quad -8 \quad 0$$

HCF

Prime factorizn:

513, 1134, 1215

$1134 = 2^1 \cdot 3^4 \cdot 7^1$

$513 = 3^3 \cdot 19^1$

$1215 = 3^5 \cdot 5^1$

HCF $= 3^3 = 27$

- $\gcd(a, b) = \gcd(b, a-b)$
- $\gcd(a, b) = \gcd(b, a \% b)$

Q. HCF of 2 #s is 18 and their sum is 324. How many such pairs of #s are possible?

→ 1st # $= 18x$ 2nd # $= 18y$

$18x + 18y = 324$ x, y will be coprime.

$x + y = 18$

sum of 2 coprimes $= 18$

1, 17 ✓	7, 11 ✓
2, 16 ✗	8, 10 ✗
3, 15 ✗	9, 9 ✗
4, 14 ✗	
5, 13 ✓	
6, 12 ✗	

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of coprimes of n that are less than/equal to $n = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$ p are prime factors of n.

$18 = 2 \cdot 3^2$

$\phi(18) = 18 \cdot \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 18 \cdot \frac{1}{2} \cdot \frac{2}{3} = 6$ ⇒ # of pairs $= 3$.

Q. HCF of 2 #s is 7 and their sum is 1470. How many such pairs of #s are possible?

$7x + 7y = 1470$

$x + y = 210$

$210 = 3 \cdot 7 \cdot 2 \cdot 5$

$210\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{7}\right) = \overset{105 \cdot 2}{210} \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} = 48$

of pairs $= \frac{48}{2} = 24$.

Q. HCF of 2 #s is 24 and their sum is 720. How many such ordered pairs of #s are possible?

→ $720 = 12 \cdot 6 \cdot 10 = 2^4 \cdot 3^2 \cdot 5$

$\overset{240}{\overset{48}{220}} \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 192$

$24x + 24y = 720$

$x + y = 30$

$30 = 2 \cdot 3 \cdot 5$

$30 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 8$

of ordered pairs $= 8$

of unordered pairs $= 4$

Q. HCF of 2 #s is 6 and their product is 30240. How many such pairs of #s are possible?

→ $6x \cdot 6y = 30240$

$xy = 840$ 2 coprimes with product 840.

$840 = 2^3 \cdot 3 \cdot 5 \cdot 7$

2^{n-1} possible ways to split as product of 2 coprimes

n: # of unique coprimes

$2^{4-1} = 8$

ordered pairs = 2^n

$= 8$

$HCF(a^m - 1, a^n - 1) = a^{HCF(m,n)} - 1$

e.g. $HCF(37^{24} - 1, 37^{21} - 1) = 37^3 - 1$

Q. $3^{192} - 1, 3^{192} + 1$

→ 3, 9, 7, 1 (any power of 3 is odd)

$3^{192} - 1$ is even.

$3^{192} + 1$ is even.

HCF = 2

HCF (consecutive even #s) = 2

HCF (consecutive odd #s) = 1

Q. HCF(7777, 777777)

→ HCF(aaa... m times, aaaa... n times) = aaaa... HCF(m, n) times

HCF(4, 6) = 2

HCF(7777, 777777) = 77

Ex. HCF(88888, 888888) = 8

7777, 777777

7(1111), 7(111111)

$7\left(\frac{9999}{9}\right), 7\left(\frac{999999}{9}\right)$

$\frac{7}{9}(10^4 - 1), \frac{7}{9}(10^6 - 1)$

$b - a = \frac{7}{9}(10^6 - 10^4)$ HCF is factor of b-a.

$= \frac{7}{9} \cdot 10^4 (10^2 - 1)$

$= \frac{7}{9} \cdot 10^4 \cdot 99$

$= 77 \cdot 10^4$

1 ← cannot be a factor for either factor for both.

LCM

For a set of numbers, it is the smallest # that is divisible by all of them.

$$HCF(x,y) \cdot LCM(x,y) = x \cdot y$$

Q. If HCF = 12, Product = 36. What is LCM?

→ $LCM = \frac{36}{12} = 3$ x Data inconsistent.

LCM cannot be less than HCF.

For fractions, $HCF = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$ $LCM = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$

$$HCF\left(\frac{2}{3}, \frac{8}{9}\right) = \frac{2}{9} \qquad LCM\left(\frac{2}{3}, \frac{8}{9}\right) = \frac{8}{3}$$

Q. $LCM(4567^{2010}+1,\ 4567^{2010}-1)$

odd — even even

7
9
3
1

HCF = 2

$$LCM = \frac{(4567^{2010})^2 - 1}{2} = \frac{4567^{4020} - 1}{2}$$

Questions

Q. Find the smallest 5-digit # divisible by 16, 24, 36, 54.

→ $16 = 2^4$, $24 = 2^3 \cdot 3$, $36 = 2^2 \cdot 3^2$, $54 = 2 \times 3^3$

$LCM = 2^4 \cdot 3^3 = 432$

$\frac{10000}{432} = 23.14$

$432 \cdot 23 = 9936$

$+ \ 432$

10368

Q. If m=500!, n=501*502*503, then the LCM of m and n is _____.
a. m b. 501m c. 502m d. None.

→ $m = 500!$

$n = 501 \times 502 \times 503$

$= 3 \times 167 \times 2 \times 251 \times 503$

(common in 5; 503 prime)

$LCM = 503\,m$

d

Q. The HCF of 2 #s is 15. How many such pairs of #s lie between 25 and 100?

→ 30 45 60 75 90

6 such pairs.

Q. The sum of 2 #s is 720 and their HCF is 24. Find the highest possible LCM.

→ $24x + 24y = 720$

$x + y = 30$ x and y are coprimes

max xy gives max LCM.

$x = 15, y = 15$ ✗

$x = 14, y = 16$ ✗

$x = 13, y = 17$

$LCM = 13 \cdot 17 \cdot 24 = 5304$.

Q. The sum of 2 #s is 1470 and their HCF is 7. Find the highest possible LCM.

→ $7x + 7y = 1470$

$x + y = 210$ x, y are coprimes.

$105, 105$ ✗

$104, 106$ ✗

$103, 107$ ✓

$103 \cdot 107 \cdot 7 = 77147$

Q. Set A is formed by selecting some of the #s from the 1st 100 natural #s such that the HCF of any 2 #s in A is 5. What is the max # of elements that set A can have.

→ 5·1 5·2 5·3 5·4 5·5 5·6 5·7 5·8 5·9 5·10 5·11 5·12 5·13 5·14 5·15 5·16 5·17 5·18 5·19 5·20

we want $5x, 5y$ s.t. x and y are coprimes.

Set A = {5, 10, 15, 25, 35, 55, 65, 85, 95}

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Q. Rahul receives a message from A, B, and C every 6, 8, and 12 minutes. If he receives all 3 at 10:30, then which of the following times can be the time at which he receives all 3 messages.
a. 12:30 p.m. b. 01:02 p.m. c. 12:48 p.m. d. None of these

→ 6, 8, 12 → LCM = 24

10:30 → all 3.
+ 24 → 10:54
+ 24 → 11:18
+ 24 → 11:42
+ 24 → 12:06
+ 24 → 12:30 ✓
+ 24 → 12:54

On dividing a number by a, b, c, if we get remainders a-k, b-k, c-k, then the # will be n*LCM(a, b, c)-k.

Q. On dividing a # by 4, 5, and 6, we get remainders 3, 4, 5. What's the #?

→ seems are 4-1, 5-1, 6-1.
LCM(4,5,6) = 60
LCM(4,5,6) − 1 = 59
= 59n.

Q. On dividing a # by 5, 6, and 7, we get remainders 3, 4, 5. What's the #?

→ LCM(5,6,7) = 42·5 = 210
210 − 2 = 208
= 208n

On dividing a number by a, b, and c, if we get remainder k, then the # will be (n-1)*LCM(a, b, c) + k.

Q. On dividing a # by 5, 6, and 7, we get remainder 2. What's the #?

→ LCM(5,6,7) = 210
(n−1)·210 + 2
210n − 208

If a # after adding k is exactly divisible by a, b, and c, then that # will be n* LCM(a, b, c) - k.

Q. Find the # that is divisible by 10, 11, 12 after adding 7 to it.

→ LCM(10,11,12) = 11·5·2·3·2 = 660
660n − 7
such #s are 553, 1313, ...

Q. LCM(1,2,3,..., 78) = n
LCM(1,2,3,..., 82) = ?

→ 79·80·81·82
prime, 2·40, 27·3, 2·41
2⁴·5 ✓, 3⁴ ?, 2·41 ✓
n doesn't have 3⁴ in there, it has 3³ at most. We need 1 more 3.
3·79·n
= 237n.

8. $\mathrm{lcm}(1, \ldots, 120) = n$

$\mathrm{lcm}(1, \ldots, 129) = ?$

121	122	123	124	125	126	127	128	129
11^2	2×61	$3 \cdot 41$	$31 \cdot 4$	5^3	$6 \cdot 21$	127	2^7 only 2^{10}	$43 \cdot 3$
11				5		127	2	

$11 \times 5 \times 127 \times 2\, n$

$= 13970\, n$